

Lab 6

Whatever Floats Your Boat



Physical Models: Maximum Buoyant Force, Weight, Volume, Density

Analysis Tools: t-score, propagation of uncertainty,

Experimental Systems: Aluminum foil boats, tupperware boats

Equipment: Weight Hangers, Precision Scales, Graduated Cylinder,

Safety Concerns: Keep the water safely away from the computers!

Introduction

In the Prelab for this week you were introduced to the concept of a buoyant force, which is the force exerted by a fluid on an object immersed in it. In particular, the *maximum* buoyant force on an object is given by:

$$F_{\text{buoyant,max}} = \rho g V_{\text{object}} \quad (6.1)$$

Where ρ is the density of the fluid, V_{object} is the volume of the object, and g is the Earth's gravitational field strength $9.81 \pm .05 \text{ N/kg}$.

This force can resist the weight force $F_{\text{weight}} = mg$. If the weight is larger than $F_{\text{buoyant,max}}$ the object will sink, otherwise it will float.

For example the density of water is $\rho = 1000 \pm 5 \text{ kg/m}^3$. We can combine the above equations to see that as long as:

$$mg \leq \rho_{\text{water}} g V_{\text{object}} \rightarrow m \leq \rho_{\text{water}} V_{\text{object}} \quad (6.2)$$

The object will float if set on water.

This means that the maximum mass a boat can carry (including its own mass) is

$$m_{max} = \rho_{water} V_{object} \quad (6.3)$$

This is the model we will be testing today.

Part 1. Floating and Sinking

Quick Check: Collect a sheet of some aluminum foil from the front of the room. Shape it into any shape you want: a boat, a ball, an origami figure, etc.

Make sure your plastic bin is filled about 2/3 full with water.

Place your aluminum shape in the water. Does it float? Does it sink?

Get another sheet, and try to make another shape which does the opposite of before (floats if it sank, sinks if it floated).

Does it seem like at least qualitatively, the relation between buoyant force and volume could explain the difference?

Designing Your Experiment: Select one of the Tupperware “boats” on your table. There should be 2 or 3 to choose from.

Then, write out a plan for a high-precision measurement to test the equation $m_{max} = \rho_{water} V_{object}$ as applied to this boat. Keep in mind:

- You’ll need to determine the volume of your object somehow. There are actually multiple ways to do this. It’s recommended you try to brainstorm at least two and then pick the one you think will give smaller uncertainty.
- You can use the precision scale to determine the mass of your boat, and of any weights you use.
- If you need to convert the readings on the graduated cylinder, recall that $1\text{ml} = 10^{-6}\text{m}^3$.
- You’ll need to propagate uncertainty. The needed partial derivatives are below.

$$\frac{\partial(\rho V)}{\partial \rho} = V \quad \frac{\partial(\rho V)}{\partial V} = \rho \quad (6.4)$$

Check in with Another Group: Briefly discuss your plans with another group. They may provide you with additional ideas and feedback. Note one similarity and one difference between your approaches.

Conducting Your Experiment: Go ahead and give it a shot. You can change your method around while you’re at it, you don’t need to treat your previous decisions as a contract.

Forming a Conclusion: Is the model consistent with your results?

Check in with Another Group: Check back in with the same group you discussed things with before. How do your results compare? Note two similarities and/or differences between what you found, and use this to inform your proposals for future iterations.

Part 2. Floating and Sinking Redux

In this part you'll proceed based on your results in Part 1:

- If you got a distinguishable result, try to diagnose why, improve your method or model, and try again using the same boat. But remeasure everything again.
- If you got an indistinguishable result, try your method on the second boat. Does it still work?

Do everything, and write everything up, as you usually would for a Part 2. We'll spare you the instructions.

Once you're done:

Bonus Challenge (Optional): Go collect some more foil from your TA (up to 12 inches in length, no more than that). Try to make a foil boat which can carry as much *added mass* as possible. Design your boat and call your TA over when you're ready to demonstrate. Whichever group gets the largest value will receive a bonus point!

Writing Up Your Lab Notes for Submission

Write up your notes for Part 1 and Part 2 in an organized way according to the rubric provided.